

Linear algebra: Exercise Chapter 6

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1. Compute determinant and inverse of the following matrices, if any.

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 6 & 0 & 2 \\ 1 & 2 & 0 \\ 4 & 1 & 1 \end{bmatrix}.$$

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2. For an $n \times n$ matrix A with $\det(A) = -1$, find:

- (a) $\det(2A^T)$;
- (b) $\det(-A)$;
- (c) $\det(A^2)$;
- (d) $\det(A^{-1})$;
- (e) $\det(C)$.

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3. Determine the determinant of the following matrices by using cofactors and selecting an appropriate row/column.

$$A = \begin{bmatrix} 8 & -11 & 7 \\ -1 & 0 & 4 \\ 7 & -3 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 3 & 0 & 3 & 0 \\ 4 & 8 & 1 & 1 \\ 7 & 0 & 1 & 0 \end{bmatrix}.$$

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4. Find the determinant of the following matrices by using determinant rules (don't compute it by using cofactors). You can use the results from the previous problem.

$$A = \begin{bmatrix} 4 & -1 & 7 \\ -9 & 2 & 2 \\ 4 & -1 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -1 & 0 \\ -9 & 2 & 0 \\ 14 & 7 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 3 & 9 \\ -1 & 0 & 4 \\ 0 & 3 & 13 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad E = \begin{bmatrix} 6 & 2 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 11 \end{bmatrix},$$

$$F = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 9 & 0 \\ 7 & 0 & -2 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 11 & 0 & 0 & 0 \\ 0 & 0 & 0 & 10 \\ 0 & 0 & -4 & 0 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & -11 & 7 \\ -1 & 0 & 4 \\ 1 & -9 & 6 \end{bmatrix}.$$

Which of the above matrices are invertible?

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5. Solve the following two systems of equations by using Cramer's rule.

$$a) \begin{cases} 12x + 3y = 15, \\ 2x - 3y = 13. \end{cases} \quad b) \begin{cases} x + y - z = 6, \\ 3x - 2y + z = -5, \\ x + 3y - 2z = 14. \end{cases}$$

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6. Let n be an odd number and A an $n \times n$ matrix such that $A^T = -A$. Show that A is not invertible.

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7. Suppose $\det(A) = 1$ and you know all the cofactors of A . How can you find A ?

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8. Write a Python code that prompts the user to enter a matrix and then compute the determinant of that.